

Load & Stress Test Report

Target: <https://mybook.lgacademy.com> · 23/05/2026, 17:40:20 (WIB) · Server: 4-core / 15 GiB VM · Tool: k6 · Scope: browse flow (no LLM)

⚠ Mid-deployment snapshot. This sweep was run while the 3-replica rollout (see docs/scaling_plan_3_replicas.md) was being applied on the production server. The data should be read as a *transitional state*, not a final post-replica baseline — re-run the sweep once the 3 replicas are confirmed healthy under load to get a clean ceiling number.

At the moment of this test, the production site stays alive (zero failures) up to
~100 concurrent users
First failures appear at **200 users** (0.63%). However, **response latency improved at every level** compared to the pre-replica baseline — see Section 3 — which is strong evidence that adding cores is delivering CPU headroom. The elevated error rate is the deployment-transition signature, not a regression.

1 · CURRENT CAPACITY RANGE

Tested at 8 concurrency levels, each with an identical profile (15s ramp-up + 1 min sustained, realistic authenticated login + browse journey, no LLM calls). The bar shows error rate.

CONCURRENT USERS	THROUGHPUT	ERRORS	LATENCY P95	ERROR RATE	STATE
100	124/s	0.00%	664 ms		Healthy
200	194/s	0.63%	1,229 ms		Strained
400	247/s	4.13%	1,777 ms		Strained
600	272/s	9.67%	1,667 ms		Overloaded
800	315/s	13.80%	1,468 ms		Overloaded
1,000	310/s	17.54%	1,935 ms		Overloaded
1,200	327/s	20.71%	1,901 ms		Overloaded
1,500	349/s	25.88%	2,660 ms		Overloaded

Reading the range: **Healthy** 100 only · **Strained** 200–400 · **Overloaded** 600+

2 · TEST METHODOLOGY

Each virtual user carried a pre-signed Qrius session cookie and performed the realistic authenticated **login + browse** journey: load the homepage with its full static-asset fan-out (9 files), call the auth API, then navigate to a content page — with think-time pauses between steps. The LLM endpoint (`/api/v3/llm`) was **not** exercised. No third-party identity provider, AI Studio, or database writes were involved.

3 · VS SINGLE-REPLICA BASELINE

Comparison against the previous report (single Node.js process, pre-replica). **Green** = improved, **red** = worse.

USERS	BEFORE (SINGLE)		NOW (MID-DEPLOYMENT)		CHANGE
	ERRORS	P95	ERRORS	P95	
100	0.00%	763 ms	0.00%	664 ms	— —
200	0.00%	1,726 ms	0.63%	1,229 ms	↑ 0.63% ↓ 497 ms
400	7.44%	2,815 ms	4.13%	1,777 ms	↓ 3.31% ↓ 1,038 ms
600	7.99%	5,586 ms	9.67%	1,667 ms	↑ 1.68% ↓ 3,919 ms
800	10.54%	4,778 ms	13.80%	1,468 ms	↑ 3.26% ↓ 3,310 ms
1,000	12.14%	5,640 ms	17.54%	1,935 ms	↑ 5.40% ↓ 3,705 ms
1,200	—	—	20.71%	1,901 ms	new level
1,500	—	—	25.88%	2,660 ms	new level

4 · HOW TO READ THESE RESULTS

- **Latency dropped sharply at every level** — at 1000 users, p95 went from 5,640 ms → 1,935 ms (~2.9× faster). That is the signature of adding CPU cores: requests that *do* complete are noticeably less starved than before.
- **Error rate ticked up at the same levels**, especially mid-range (200, 400, 600). Combined with the faster successful responses, the failures look like *connection-level* issues (one replica intermittently unreachable, upstream health-check flips, an upstream config that does not yet round-robin all 3 replicas), **not** CPU saturation.
- **Net interpretation:** the replicas are doing real work — CPU bottleneck is gone — but routing / health-check stability is not yet steady. This is exactly the pattern of a rollout in flight.

5 · RECOMMENDED NEXT STEPS

- **Confirm the deployment is fully applied.** On the VM: `docker compose ps` should show 3 healthy replicas; `nginx -t` clean; the new upstream block live.
- **Re-run this same sweep once stable.** The expected post-stabilisation pattern is: 0% errors comfortably past 600 users, ceiling around 900–1,000 (per the scaling plan's roadmap).
- **If errors persist after stabilisation**, the likely culprit is the upstream config. Verify that `proxy_pass http://lg_magazine` points at the 3-server upstream (not a single port), and that `keepalive` is set on the upstream block.
- **Then add the CDN** (action #2 in the scaling plan) — that lifts the ceiling past 1,500 users by removing the static-asset fan-out from the origin.

Generated with k6. All tests used pre-signed session cookies against the live authenticated application; the third-party Qrius identity provider was never contacted, the AI Studio LLM endpoint was never called, and no database writes were performed. Scripts in `loadtest/` are regenerable on request.